CS 6012: Autumn 2026 — Open Project

Intermediate Checkpoint #4: Status Report (Submission 2: Code-Based Project)

Team: Aayush Gupta (25m0140) & Srijal Gupta (25m0387) **Date:** October 24, 2026

1. Title of the Project

GNN Molecular Substructure Search: Accuracy vs. Speed Trade-offs Against Classical Fingerprint Screening

2. Updates on Problem Statement (Two-Stage Cascaded Screening)

In large chemical libraries (50,000+ molecules), standalone neural classification cannot guarantee zero false negatives. We have formally adapted our research formulation into a **two-stage cascaded screening architecture**: (1) A learned order-embedding GNN ($z_Q \leq z_T$) rapidly screens the whole database to shortlist candidate molecules at a strict high-recall threshold (≥98%), and (2) RDKit's exact matcher (HasSubstructMatch) verifies *only* the shortlist, yielding 100% precision. Our empirical focus is quantifying the speedup gained over brute force exact matching while benchmarking directly against the industrial baseline: RDKit's 2048-bit Chem.PatternFingerprint screen followed by exact verification.

3. Difficulties Encountered & Technical Insights

- **Shortlist Size & Verification Bottleneck:** While GNN B's order-embedding vector check is extremely fast on GPU (2.79 ms over 5,000 molecules), its lower precision at the ≥98% recall point returns an average of 1,414 candidates (vs. only 703 for Pattern Fingerprints). Because RDKit verification takes ~0.05 ms/mol, verifying those extra false positives increases overall cascaded latency to 20.2 ms (compared to 5.6 ms for Pattern FP + RDKit). Tighter precision calibration is essential to keep verification overhead from dominating.
- **Query Size Degradation (4 vs. 12 Atoms):** Empirical breakdown shows strong recall (>83%) on small 4–5 atom subgraphs, but recall drops to 28%–44% on larger 9–12 atom queries. The 4-layer GINE receptive field and global sum-pooling struggle to preserve strict subgraph isomorphism as molecular diameter and cycle complexity expand.
- **Hard Negative Mining Bottleneck:** Training order embeddings requires hard negatives (pairs passing fingerprint screens but failing exact matching). Generating these requires intensive RDKit exact calls during batch construction, causing CPU-side data preprocessing bottlenecks when scaling beyond 40,000 pairs.

4. Contributions of Members

- **Aayush Gupta (25m0140):** GPU acceleration and CUDA runtime integration on RTX 3050, PyG GINEConv architecture tuning, order-embedding max-margin loss formulation, high-recall threshold calibration, and Checkpoint documentation.
- **Srijal Gupta (25m0387):** ZINC250k query generation stratified across atom sizes (4–12 atoms + 20 functional groups), hard-negative sampling pipeline, classical Pattern Fingerprint benchmarking suite, error analysis by query size, and schedule tracking.
- **Joint Deliverables:** Cascaded pipeline evaluation, error analysis documentation, presentation slide preparation, and GitHub maintenance.

5. GitHub Submission & Updated Schedule

- **GitHub Repository:** https://github.com/srijal20/GNN-Substructure-Search-project
  (Active commits by both members; contains full source code gnn_substructure_search.py, Colab notebook, pilot CSV benchmarks, and tradeoff plots)
- **Updated Schedule Spreadsheet:** Submitted as companion PDF (Updated_Schedule_Spreadsheet.pdf) and online at: Google Spreadsheet Link